MODULE 4: ORDINARY DIFFERENTIAL EQUATIONS (CO: 4)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BAMAT101 Official Syllabus, VIT Department of Mathematics Question Bank, and VIT FAT Past Papers

Scope of This Textbook (All 8 Syllabus Topics): This textbook covers every topic in Module 4 of your syllabus screenshot: (1) Introduction to linear second-order ODEs with constant coefficients, (2) Solutions of ODE by method of undetermined coefficients, (3) Solutions of ODE by method of variation of parameters–1, (4) Solutions of ODE by method of variation of parameters–2, (5) Solution of Cauchy-Euler differential equation, (6) Solution of Cauchy-Legendre differential equations, (7) Steady-state response of input signal in digital signal processing, and (8) Differential equations for LCR circuit simulation.

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1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 4

In VIT Final Assessment Tests (BMAT102L / BAMAT101 FAT Papers), **Module 4 carries 20 to 30 marks** across two or three 10-mark questions:

1. **Question Type A (10 Marks)**: Solving a constant-coefficient ODE $(aD^2+bD+c)y = R(x)$ by the **Method of Undetermined Coefficients** (especially when $R(x) = e^{ax}\cos bx$, $x^n e^{ax}$, or when the Duplication Rule x^s is triggered).
2. **Question Type B (10 Marks)**: Solving either a trigonometric ODE like $(D^2+a^2)y = \tan ax, \sec ax, \csc ax$ or a **Cauchy-Euler / Cauchy-Legendre ODE** like $(x^2D^2 - xD + 1)y = x\log x$ or $(x^2D^2 - 4xD + 6)y = \sin(\log x)$ using the **Method of Variation of Parameters**.
3. **Question Type C (10 Marks)**: Formulating and solving an **LCR series circuit** or **damped forced spring-mass system** and interpreting the transient vs. steady-state response.

Table 1: Complementary Function (y_c) & Trial Particular Integral (y_p) Master Reference

Auxiliary Roots of $am^2+bm+c=0$	**Complementary Function** $y_c(x)$	**Cauchy-Euler Equivalent** ($x = e^t$, $\theta = \frac{d}{dt}$)
Distinct Real: $m_1 \neq m_2$	$C_1e^{m_1x} + C_2e^{m_2x}$	$C_1x^{m_1} + C_2x^{m_2}$
Repeated Real: $m_1 = m_2 = m$	$(C_1 + C_2x)e^{mx}$	$(C_1 + C_2\ln x)x^m$
Complex Conjugate: $\alpha \pm i\beta$	$e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x)$	$x^\alpha\big[C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x)\big]$

2 Topic 1: Linear Second-Order ODEs with Constant Coefficients

2.1 Theory: Homogeneous Equations, Initial Value Problems & Harmonic Motion

A linear second-order ODE with constant coefficients has the form:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = R(x) \iff (aD^2+bD+c)y = R(x), \qquad \text{where } D = \frac{d}{dx}.$$

The general solution is $\boxed{y(x) = y_c(x) + y_p(x)}$, where $y_c(x)$ is the **Complementary Function (CF)** solving $(aD^2+bD+c)y = 0$, obtained by substituting $y = e^{mx}$ to get the **Auxiliary Equation (AE)** $am^2+bm+c=0$.

2.2 Worked VIT Question Bank & Tutorial Problems (Topic 1)

Problem 4.1.1 (VIT Question Bank Q1, Q2, Q3 & Q4): *Solve the following homogeneous ODEs and initial/boundary value problems: (a)* $\frac{d^2x}{dt^2}+3\frac{dx}{dt}+2x=0$ *(b)* $\frac{d^2y}{dx^2}-10\frac{dy}{dx}+25y=0$ *(c)* $l\frac{d^2\theta}{dt^2}+g\theta=0$ *with* $\theta(0)=\alpha, \theta'(0)=0$ *(d)* $\frac{d^2x}{dt^2}+4x=0$ *with* $x(0)=3, \frac{dx}{dt}(0)=8$. *Also find the maximum value of* $x(t)$.

Solution:

Part (a): AE is $m^2+3m+2=(m+1)(m+2)=0 \implies m=-1,-2$. Thus $x(t) = C_1e^{-t} + C_2e^{-2t}$.

Part (b): AE is $m^2-10m+25=(m-5)^2=0 \implies m=5,5$ (repeated). Thus $y(x) = (C_1+C_2x)e^{5x}$.

Part (c): Dividing by l: $\theta'' + \frac{g}{l}\theta = 0$. AE is $m^2+\frac{g}{l}=0 \implies m = \pm i\sqrt{\frac{g}{l}}$. Thus $\theta(t) = C_1\cos\left(\sqrt{\frac{g}{l}}\,t\right) + C_2\sin\left(\sqrt{\frac{g}{l}}\,t\right)$. Applying $\theta(0)=\alpha \implies C_1=\alpha$, and $\theta'(0)=0 \implies C_2=0$, so $\theta(t) = \alpha\cos\left(\sqrt{\frac{g}{l}}\,t\right)$.

Part (d): AE is $m^2+4=0 \implies m=\pm 2i$, so $x(t)=C_1\cos(2t)+C_2\sin(2t)$. At $t=0$: $x(0)=C_1=3$, and $x'(0)=2C_2=8 \implies C_2=4$. Hence $x(t)=3\cos(2t)+4\sin(2t)$. The maximum amplitude of $A\cos\omega t+B\sin\omega t$ is $\sqrt{A^2+B^2}=\sqrt{3^2+4^2}=5$. □

3 Topic 2: Solutions of ODE by Method of Undetermined Coefficients

3.1 Theory & The Duplication (Resonance) Rule

When $R(x)$ consists of polynomials, exponentials e^{ax}, sines/cosines $\sin bx, \cos bx$, or sums and products of these, we assume a trial form for $y_p(x)$ with unknown constants $A, B, \ldots$:

Table 2: Method of Undetermined Coefficients Trial Table & Duplication Rule

Term in $R(x)$	**Base Trial Form $y_p(x)$**	**Duplication / Modification Rule (CRITICAL)**
ke^{ax}	Ae^{ax}	If a is a single root of AE, multiply by x: Axe^{ax}. If a is a double root of AE, multiply by x^2: Ax^2e^{ax}.
$k\cos(bx)$ or $k\sin(bx)$	$A\cos(bx)+B\sin(bx)$	If $\pm ib$ are roots of AE, multiply by x: $x(A\cos bx+B\sin bx)$.
Polynomial $P_n(x)$ of deg n	$A_nx^n+\cdots+A_1x+A_0$	If $m=0$ is a root of AE of multiplicity s, multiply by x^s.
$e^{ax}\cos(bx)$ or $e^{ax}\sin(bx)$	$e^{ax}(A\cos bx+B\sin bx)$	If $a\pm ib$ are roots of AE, multiply by x.
$P_n(x)e^{ax}$	$(A_nx^n+\cdots+A_0)e^{ax}$	If a is a root of AE of multiplicity s, multiply the entire polynomial by x^s.

3.2 Worked VIT Question Bank & Past FAT Problems (Topic 2)

Problem 4.2.1 (BMAT102L FAT April 2025 Slot A1 Q1 — 10 Marks): *Solve $(D^2-4D+3)y=e^x\cos 2x$ by the method of undetermined coefficients.*

Solution:

Step 1 (Complementary Function y_c): The auxiliary equation is $m^2-4m+3=(m-1)(m-3)=0 \implies m=1,3$.

$$y_c(x)=C_1e^x+C_2e^{3x}.$$

Step 2 (Trial Particular Integral y_p): Here $R(x)=e^x\cos 2x$, which corresponds to complex root $1\pm 2i$. Since $1\pm 2i$ is **not** a root of the AE (even though 1 is a root, $1\pm 2i\neq 1$), no x multiplier is needed! Assume:

$$y_p(x)=e^x\left(A\cos 2x+B\sin 2x\right).$$

Differentiating using the product rule:

$$y_p'(x)=e^x\Big[(A+2B)\cos 2x+(B-2A)\sin 2x\Big],$$
$$y_p''(x)=e^x\Big[(-3A+4B)\cos 2x+(-3B-4A)\sin 2x\Big].$$

Substitute y_p, y_p', y_p'' into $y_p''-4y_p'+3y_p=e^x\cos 2x$:

$$e^x\Big[(-3A+4B-4A-8B+3A)\cos 2x+(-3B-4A-4B+8A+3B)\sin 2x\Big]=e^x\cos 2x,$$

$$(-4A-4B)\cos 2x + (4A-4B)\sin 2x = 1\cdot\cos 2x + 0\cdot\sin 2x.$$

Equating coefficients of $\cos 2x$ and $\sin 2x$:

$$4A-4B=0 \implies A=B, \qquad -4A-4B=1 \implies -8A=1 \implies A=B=-\frac{1}{8}.$$

Hence $y_p(x) = -\frac{1}{8}e^x(\cos 2x + \sin 2x)$, and the general solution is:

$$y(x) = C_1e^x + C_2e^{3x} - \frac{1}{8}e^x(\cos 2x + \sin 2x). \quad \square$$

Problem 4.2.2 (VIT Question Bank Q5 & Q12 — Duplication Rule in Action): *Solve by the method of undetermined coefficients: (a)* $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x} + \cos 2x$ *and (b)* $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^3e^{2x} + xe^{2x}$.

Solution:

In both parts, the AE is $m^2-4m+4 = (m-2)^2 = 0 \implies m = 2, 2$ (double root!), so $y_c(x) = (C_1+C_2x)e^{2x}$.

Part (a): For $R_1(x) = e^{2x}$, since $m = 2$ is a **double root** of the AE, we multiply the base trial Ae^{2x} by x^2, giving $y_{p1} = Ax^2e^{2x}$. For $R_2(x) = \cos 2x$, $\pm 2i$ is not a root, so $y_{p2} = B\cos 2x + C\sin 2x$.

- Substituting $y_{p1} = Ax^2e^{2x}$ into $(D-2)^2y_{p1} = e^{2x}$ gives $2Ae^{2x} = e^{2x} \implies A = \frac{1}{2}$.
- Substituting $y_{p2} = B\cos 2x + C\sin 2x$ into $y'' - 4y' + 4y = \cos 2x$ gives $-8C\cos 2x + 8B\sin 2x = \cos 2x \implies B = 0, C = -\frac{1}{8}$.

Thus $y(x) = (C_1 + C_2x)e^{2x} + \frac{1}{2}x^2e^{2x} - \frac{1}{8}\sin 2x$.

Part (b): For $R(x) = (x^3 + x)e^{2x}$, since $m = 2$ is a double root of AE, we multiply the cubic trial $(Ax^3 + Bx^2 + Cx + D)e^{2x}$ by x^2:

$$y_p(x) = x^2(Ax^3 + Bx^2 + Cx + D)e^{2x} = (Ax^5 + Bx^4 + Cx^3 + Dx^2)e^{2x}.$$

Since $(D-2)^2[g(x)e^{2x}] = g''(x)e^{2x}$, we immediately have:

$$g''(x) = 20Ax^3 + 12Bx^2 + 6Cx + 2D = x^3 + x \implies A = \frac{1}{20},\ B = 0,\ C = \frac{1}{6},\ D = 0.$$

Thus $y(x) = (C_1 + C_2x)e^{2x} + \left(\frac{1}{20}x^5 + \frac{1}{6}x^3\right)e^{2x}$. $\square$

Problem 4.2.3 (VIT Question Bank Q20): *Solve* $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3y = x^3 + \sin x$ *by the method of undetermined coefficients.*

Solution:

1. **Complementary Function**: $m^2-2m+3 = 0 \implies m = 1\pm i\sqrt{2}$, so $y_c(x) = e^x\left(C_1\cos(\sqrt{2}x) + C_2\sin(\sqrt{2}x)\right)$.
2. **Trial Particular Integral**: Assume $y_p(x) = Ax^3 + Bx^2 + Cx + D + E\cos x + F\sin x$. Substituting into $y'' - 2y' + 3y = x^3 + \sin x$:

- **Polynomial part**: $3Ax^3 + (3B-6A)x^2 + (6A-4B+3C)x + (2B-2C+3D) = x^3 \implies A = \frac{1}{3}, B = \frac{2}{3}, C = \frac{2}{9}, D = -\frac{8}{27}$.
- **Trigonometric part**: $(2E-2F)\cos x + (2E+2F)\sin x = \sin x \implies E = \frac{1}{4}, F = \frac{1}{4}$.

Combining $y = y_c + y_p$ gives the complete solution. $\square$

4 Topic 3: Method of Variation of Parameters – 1 (Constant Coefficients)

4.1 Theory & Wronskian Formula

To solve a linear second-order ODE in **standard form** (where the coefficient of y'' is strictly 1):

$$\boxed{y'' + P(x)y' + Q(x)y = R(x)}$$

given the complementary function $y_c(x) = C_1 y_1(x) + C_2 y_2(x)$:

1. Compute the **Wronskian** $W(y_1, y_2)$:
$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0.$$
2. The particular integral is $\boxed{y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)}$, where:
$$\boxed{u_1(x) = -\int \frac{y_2(x)\,R(x)}{W}\,dx \qquad \text{and} \qquad u_2(x) = \int \frac{y_1(x)\,R(x)}{W}\,dx.}$$

4.2 Worked VIT Question Bank & Tutorial Problems (Topic 3)

Problem 4.3.1 (VIT Question Bank Q11): *Apply the method of variation of parameters to solve* $(D^2 + 4)y = 4\tan 2x$.

Solution:

1. **Complementary Function**: $m^2 + 4 = 0 \implies m = \pm 2i$, so $y_c = C_1 \cos 2x + C_2 \sin 2x$. Here $y_1 = \cos 2x$, $y_2 = \sin 2x$, and $R(x) = 4\tan 2x$. 2. **Wronskian** W:

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2(\cos^2 2x + \sin^2 2x) = 2.$$

3. **Compute** $u_1(x)$ **and** $u_2(x)$:

$$u_1(x) = -2\int \frac{\sin^2 2x}{\cos 2x}\,dx = -2\int (\sec 2x - \cos 2x)\,dx = -\ln|\sec 2x + \tan 2x| + \sin 2x,$$
$$u_2(x) = \int \frac{\cos 2x \cdot 4\tan 2x}{2}\,dx = 2\int \sin 2x\,dx = -\cos 2x.$$

4. **Particular Integral** $y_p = u_1 y_1 + u_2 y_2$:

$$y_p = (-\ln|\sec 2x + \tan 2x| + \sin 2x)\cos 2x + (-\cos 2x)\sin 2x = -\cos 2x \ln|\sec 2x + \tan 2x|.$$

Thus $y(x) = C_1 \cos 2x + C_2 \sin 2x - \cos 2x \ln|\sec 2x + \tan 2x|$. □

Problem 4.3.2 (VIT Question Bank Q13, Q21 & Tutorial Q7): *Solve by the method of variation of parameters: (a)* $(D^2 + 1)y = \csc x$ *(b)* $(D^2 + 1)y = \sec x$ *(c)* $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x \sin x$.

Solution:

Part (a): $y_1 = \cos x, y_2 = \sin x, W = 1, R(x) = \csc x$.

$$u_1 = -\int \sin x \csc x\,dx = -x, \qquad u_2 = \int \cos x \csc x\,dx = \int \cot x\,dx = \ln|\sin x|.$$

Thus $y(x) = C_1 \cos x + C_2 \sin x - x\cos x + \sin x \ln|\sin x|$.

Part (b): $y_1 = \cos x, y_2 = \sin x, W = 1, R(x) = \sec x$.

$$u_1 = -\int \sin x \sec x \, dx = \ln|\cos x|, \qquad u_2 = \int \cos x \sec x \, dx = x.$$

Thus $y(x) = C_1 \cos x + C_2 \sin x + \cos x \ln|\cos x| + x \sin x$.

Part (c): AE is $m^2 - 2m = 0 \implies m = 0, 2$, so $y_1 = 1$, $y_2 = e^{2x}$, $W = 2e^{2x}$, and $R(x) = e^x \sin x$.

$$u_1 = -\int \frac{e^{2x} \cdot e^x \sin x}{2e^{2x}} \, dx = -\frac{1}{2}\int e^x \sin x \, dx = -\frac{1}{4}e^x(\sin x - \cos x),$$

$$u_2 = \int \frac{1 \cdot e^x \sin x}{2e^{2x}} \, dx = \frac{1}{2}\int e^{-x} \sin x \, dx = -\frac{1}{4}e^{-x}(\sin x + \cos x).$$

Then $y_p = u_1(1) + u_2(e^{2x}) = -\frac{1}{4}e^x(\sin x - \cos x) - \frac{1}{4}e^x(\sin x + \cos x) = -\frac{1}{2}e^x \sin x$. □

5 Topic 4: Method of Variation of Parameters – 2 (Variable Coefficients)

[VIT FAT EXAM INSIGHT] The #1 Trap in VIT FAT Papers on Variation of Parameters: In VIT FAT April 2025, the exam asks: *"Solve $(x^2D^2 - xD + 1)y = x \log x$ by the method of variation of parameters"* or *"Solve $(x^2D^2 - 4xD + 6)y = \sin(\log x)$ by the method of variation of parameters."* **CRITICAL RULE**: Before computing $u_1 = -\int \frac{y_2 R(x)}{W} dx$ and $u_2 = \int \frac{y_1 R(x)}{W} dx$, you **MUST divide the entire ODE by the leading coefficient** $a(x) = x^2$ so the coefficient of y'' is 1! That is, $R(x) = \frac{\text{RHS}}{x^2}$ (OR transform the entire ODE using $x = e^t$ so it has constant coefficients with respect to t and apply variation of parameters in t). Let us show both!

Problem 4.4.1 (BMAT102L FAT April 2025 Slot A1 Q2 — 10 Marks): *Solve $(x^2D^2 - xD + 1)y = x \ln x$ by the method of variation of parameters.*

Solution:

Step 1 (Divide by x^2 to put in Standard Form $y'' + P(x)y' + Q(x)y = R(x)$):

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = \frac{\ln x}{x}, \qquad \text{so } R(x) = \frac{\ln x}{x}.$$

Step 2 (Find y_c via Cauchy-Euler substitution $y = x^m$): Substituting $y = x^m$ into $x^2y'' - xy' + y = 0$ gives $m(m-1) - m + 1 = (m-1)^2 = 0 \implies m = 1, 1$. Thus $y_c(x) = C_1 x + C_2 x \ln x$, with $y_1(x) = x$ and $y_2(x) = x \ln x$. **Step 3 (Compute Wronskian $W(y_1, y_2)$):**

$$W = \begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} = x(1 + \ln x) - x \ln x = x.$$

Step 4 (Compute $u_1(x)$ and $u_2(x)$ using $R(x) = \frac{\ln x}{x}$):

$$u_1(x) = -\int \frac{y_2(x)R(x)}{W} dx = -\int \frac{(x \ln x)\left(\frac{\ln x}{x}\right)}{x} dx = -\int \frac{(\ln x)^2}{x} dx = -\frac{(\ln x)^3}{3},$$

$$u_2(x) = \int \frac{y_1(x)R(x)}{W} dx = \int \frac{x\left(\frac{\ln x}{x}\right)}{x} dx = \int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2}.$$

Step 5 (Construct y_p and General Solution):

$$y_p(x) = u_1 y_1 + u_2 y_2 = -\frac{(\ln x)^3}{3}x + \frac{(\ln x)^2}{2}(x \ln x) = \frac{1}{6}x(\ln x)^3.$$

$$\boxed{y(x) = C_1 x + C_2 x \ln x + \frac{1}{6} x (\ln x)^3.} \quad \square$$

Problem 4.4.2 (BMAT102L FAT April 2025 Paper 2 — 10 Marks): *Solve* $(x^2D^2 - 4xD + 6)y = \sin(\ln x)$ *by the method of variation of parameters.*

Solution:

1. **Standard Form**: Divide by x^2: $y'' - \frac{4}{x}y' + \frac{6}{x^2}y = \frac{\sin(\ln x)}{x^2}$, so $R(x) = \frac{\sin(\ln x)}{x^2}$. 2. **Complementary Function**: $m(m-1) - 4m + 6 = m^2 - 5m + 6 = (m-2)(m-3) = 0 \implies m = 2, 3$. Thus $y_1 = x^2$, $y_2 = x^3$, and $W = \begin{vmatrix} x^2 & x^3 \\ 2x & 3x^2 \end{vmatrix} = x^4$. 3. **Compute $u_1(x)$ and $u_2(x)$** (using substitution $t = \ln x \implies dt = \frac{dx}{x}$):

$$u_1(x) = -\int x^{-2} \sin(\ln x) \frac{dx}{x} = -\int e^{-2t} \sin t \, dt = \frac{2\sin(\ln x) + \cos(\ln x)}{5x^2},$$
$$u_2(x) = \int x^{-3} \sin(\ln x) \frac{dx}{x} = \int e^{-3t} \sin t \, dt = -\frac{3\sin(\ln x) + \cos(\ln x)}{10x^3}.$$

4. **Combine** $y_p = u_1 x^2 + u_2 x^3$:

$$y_p(x) = \frac{4\sin(\ln x) + 2\cos(\ln x) - 3\sin(\ln x) - \cos(\ln x)}{10} = \frac{\sin(\ln x) + \cos(\ln x)}{10}.$$

Thus $y(x) = C_1 x^2 + C_2 x^3 + \frac{1}{10}[\sin(\ln x) + \cos(\ln x)]$. $\square$

6 Topic 5 & Topic 6: Cauchy-Euler and Cauchy-Legendre Differential Equations

6.1 Theory: Operator Transformations to Constant Coefficients

1. **Cauchy-Euler Equation** $(ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = R(x))$: Substitute $\boxed{x = e^t \iff t = \ln x}$, and define $\theta = \frac{d}{dt}$. Then:

$$\boxed{xDy = \theta y, \qquad x^2D^2y = \theta(\theta-1)y, \qquad x^3D^3y = \theta(\theta-1)(\theta-2)y.}$$

2. **Cauchy-Legendre Equation** $(a_0(ax+b)^2\frac{d^2y}{dx^2} + a_1(ax+b)\frac{dy}{dx} + a_2y = R(x))$: Substitute $\boxed{ax + b = e^t \iff t = \ln(ax+b)}$, and define $\theta = \frac{d}{dt}$. By chain rule:

$$\boxed{(ax+b)Dy = a\theta y, \qquad (ax+b)^2D^2y = a^2\theta(\theta-1)y.}$$

6.2 Worked VIT Question Bank Problems (Topics 5 & 6)

Problem 4.5.1 (VIT Question Bank Q6, Q8, Q10 & Q19): *Solve the following Cauchy-Euler differential equations: (a)* $2x^2y'' + 3xy' - 3y = x^3$ *(b)* $t^2\frac{d^2x}{dt^2} + t\frac{dx}{dt} + 9x = \ln t$ *(c)* $(x^2D^2 - 3xD + 5)y = \sin(\ln x)$.

Solution:

Part (a): Let $x = e^t$ $(\theta = \frac{d}{dt})$: $[2\theta(\theta-1) + 3\theta - 3]y = (2\theta^2 + \theta - 3)y = e^{3t}$. AE: $2m^2 + m - 3 = (2m+3)(m-1) = 0 \implies m = 1, -3/2$, so $y_c = C_1x + C_2x^{-3/2}$. For y_p: $\frac{1}{2\theta^2+\theta-3}e^{3t} = \frac{e^{3t}}{2(3)^2+3-3} = \frac{e^{3t}}{18} = \frac{x^3}{18}$. Thus $y(x) = C_1x + C_2x^{-3/2} + \frac{x^3}{18}$.

Part (b): Let $t = e^u$ $(\theta = \frac{d}{du})$: $[\theta(\theta-1) + \theta + 9]x = (\theta^2 + 9)x = u$. AE: $m^2 + 9 = 0 \implies m = \pm 3i \implies x_c = A\cos(3\ln t) + B\sin(3\ln t)$. Particular solution: $x_p = \frac{1}{9+\theta^2}u = \frac{u}{9} = \frac{\ln t}{9}$. Thus $x(t) = A\cos(3\ln t) + B\sin(3\ln t) + \frac{\ln t}{9}$.

Part (c): Let $x = e^t$ ($\theta = \frac{d}{dt}$): $[\theta(\theta-1)-3\theta+5]y = (\theta^2-4\theta+5)y = \sin t$. AE: $m^2-4m+5=0 \implies m = 2\pm i \implies y_c = x^2[A\cos(\ln x)+B\sin(\ln x)]$. For y_p: Assuming $y_p = C\cos t + D\sin t$ in $y''-4y'+5y = \sin t$ gives $(4C-4D)\cos t + (4C+4D)\sin t = \sin t \implies C = D = \frac{1}{8}$, so $y_p = \frac{1}{8}[\cos(\ln x)+\sin(\ln x)]$. *(Note: If the equation is $(x^2D^2-3xD+5)y = x^2\sin(\ln x) = e^{2t}\sin t$, which is the resonance case for $m = 2\pm i$, then $y_p = -\frac{t}{2}e^{2t}\cos t = -\frac{x^2\ln x}{2}\cos(\ln x)$, matching VIT Question Bank Q19!)* □

Problem 4.6.1 (VIT FAT Classic — Cauchy-Legendre Equation): *Solve the Cauchy-Legendre differential equation* $(2x+3)^2\frac{d^2y}{dx^2} - 2(2x+3)\frac{dy}{dx} - 12y = 6x$.

Solution:

Let $2x+3 = e^t \implies t = \ln(2x+3)$ and $x = \frac{e^t-3}{2}$, so $6x = 3e^t - 9$. Using $(2x+3)D = 2\theta$ and $(2x+3)^2D^2 = 4\theta(\theta-1)$:

$$\Big[4\theta(\theta-1) - 2(2\theta) - 12\Big]y = \Big(4\theta^2 - 8\theta - 12\Big)\,y = 3e^t - 9.$$

1. **Complementary Function**: $4m^2-8m-12 = 4(m-3)(m+1) = 0 \implies m = 3, -1$.

$$y_c = C_1e^{3t} + C_2e^{-t} = C_1(2x+3)^3 + C_2(2x+3)^{-1}.$$

2. **Particular Integral**:

$$y_p = \frac{3e^t}{4(1)^2-8(1)-12} - \frac{9e^{0t}}{4(0)-8(0)-12} = -\frac{3}{16}e^t + \frac{9}{12} = -\frac{3}{16}(2x+3) + \frac{3}{4}. \quad \square$$

7 Topic 7 & Topic 8: Steady-State DSP Response & LCR Circuit Simulation

7.1 Theory: LCR Series Circuits, Damping Regimes & Steady-State Signal Response

By Kirchhoff's Voltage Law (KVL), an inductor L (Henries), resistor R (Ohms), and capacitor C (Farads) connected in series with an electromotive force $E(t)$ (Volts) satisfy:

$$\boxed{L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t), \qquad \text{where current } i(t) = \frac{dq}{dt}.}$$

Similarly, a damped spring-mass mechanical oscillator (or continuous-time DSP filter) satisfies $m\frac{d^2y}{dt^2} + c\frac{dy}{dt} + ky = F_0\cos(\omega t)$.

- **Transient Response** $q_{\mathbf{tr}}(t) = q_c(t)$: Since $R, L, C > 0$, the roots of $Lm^2 + Rm + \frac{1}{C} = 0$ have negative real parts, so $q_c(t) \to 0$ exponentially as $t \to \infty$ (*overdamped* if $R^2 > 4L/C$, *critically damped* if $R^2 = 4L/C$, *underdamped* if $R^2 < 4L/C$).
- **Steady-State Response** $q_{\mathbf{ss}}(t) = q_p(t)$: The particular integral $q_p(t)$ persists as $t \to \infty$. For sinusoidal input $E(t) = E_0\sin(\omega t)$ or $E_0\cos(\omega t)$, the steady-state current has frequency ω with amplitude $\frac{E_0}{\sqrt{R^2+(\omega L - 1/(\omega C))^2}}$.

7.2 Worked VIT Question Bank & FAT Problems (Topics 7 & 8)

Problem 4.8.1 (VIT Question Bank Q15 — Full LCR Series Circuit Problem): *An inductor of* $L = 2$ *H, resistor of* $R = 16\,\Omega$*, and capacitor of* $C = 0.02$ *F are connected in series with a battery of EMF* $E(t) = 100\sin 2t$*. At* $t = 0$*, the charge on the capacitor and current in the circuit are zero* ($q(0) = 0, i(0) = 0$)*. Find the charge* $q(t)$ *and current* $i(t)$ *for* $t > 0$.

Solution:

1. **Formulate the ODE**: Substituting $L = 2, R = 16, \frac{1}{C} = \frac{1}{0.02} = 50$, and $E(t) = 100\sin 2t$:

$$2\frac{d^2q}{dt^2} + 16\frac{dq}{dt} + 50q = 100\sin 2t \implies \frac{d^2q}{dt^2} + 8\frac{dq}{dt} + 25q = 50\sin 2t.$$

2. **Transient (Complementary) Solution** $q_c(t)$: AE: $m^2+8m+25 = 0 \implies m = \frac{-8\pm\sqrt{64-100}}{2} = -4\pm 3i$ (underdamped).

$$q_c(t) = e^{-4t}\left(C_1\cos 3t + C_2\sin 3t\right).$$

3. **Steady-State (Particular) Solution** $q_p(t)$: Assume $q_p(t) = A\cos 2t + B\sin 2t$. Substituting into $q'' + 8q' + 25q = 50\sin 2t$:

$$(21A + 16B)\cos 2t + (21B - 16A)\sin 2t = 50\sin 2t \implies 21A + 16B = 0,\ \ -16A + 21B = 50.$$

Solving gives $A = -\frac{160}{89}$ and $B = \frac{210}{89}$ (or if $E = 50\sin 2t$, $B = \frac{105}{89}$). 4. **Apply Initial Conditions** $q(0) = 0, q'(0) = 0$:

- $q(0) = C_1 - \frac{160}{89} = 0 \implies C_1 = \frac{160}{89}$.
- $q'(0) = -4C_1 + 3C_2 + 2B = 0 \implies 3C_2 = 4\left(\frac{160}{89}\right) - 2\left(\frac{210}{89}\right) = \frac{220}{89} \implies C_2 = \frac{220}{267}$.

Thus:

$$q(t) = \underbrace{e^{-4t}\left(\frac{160}{89}\cos 3t + \frac{220}{267}\sin 3t\right)}_{\text{Transient response } q_{\text{tr}}(t)\to 0} + \underbrace{\left(-\frac{160}{89}\cos 2t + \frac{210}{89}\sin 2t\right)}_{\text{Steady-state response } q_{\text{ss}}(t)}, \qquad \text{and } i(t) = \frac{dq}{dt}. \quad \square$$

Problem 4.8.2 (VIT Question Bank Q17 & BMAT102L FAT April 2025): *In an LCR circuit, the charge $q(t)$ on a capacitor plate satisfies $\frac{d^2q}{dt^2} + 2\frac{dq}{dt} + q = 2\sin 2t$. Find $q(t)$ if initially $q(0) = 0$ and $i(0) = q'(0) = 0$, and interpret the result physically.*

Solution:

1. **Transient part** $q_c(t)$: AE is $m^2 + 2m + 1 = (m+1)^2 = 0 \implies m = -1, -1$ (**critically damped**).

$$q_c(t) = (C_1 + C_2 t)e^{-t}.$$

2. **Steady-state part** $q_p(t)$: Substitute $q_p = A\cos 2t + B\sin 2t$ into $q'' + 2q' + q = 2\sin 2t$:

$$(-3A + 4B)\cos 2t + (-3B - 4A)\sin 2t = 2\sin 2t \implies A = -\frac{8}{25},\ \ B = -\frac{6}{25}.$$

3. **Apply** $q(0) = 0, q'(0) = 0$:

$$q(0) = C_1 - \frac{8}{25} = 0 \implies C_1 = \frac{8}{25}, \qquad q'(0) = -C_1 + C_2 + 2B = -\frac{8}{25} + C_2 - \frac{12}{25} = 0 \implies C_2 = \frac{20}{25} = \frac{4}{5}.$$

$$\boxed{q(t) = \left(\frac{8}{25} + \frac{4}{5}t\right)e^{-t} - \frac{8}{25}\cos 2t - \frac{6}{25}\sin 2t.}$$

Physical Interpretation: The first term $\left(\frac{8}{25} + \frac{4}{5}t\right)e^{-t}$ is the **critically damped transient charge**, which decays rapidly to zero as $t \to \infty$. The remaining terms $-\frac{8}{25}\cos 2t - \frac{6}{25}\sin 2t = -\frac{2}{5}\sin(2t + \phi)$ represent the **steady-state sinusoidal oscillation** at the driving frequency $\omega = 2$ rad/s with constant amplitude $\frac{2}{5}$ C. $\square$